

IPW3 : ONERAM6 Code-to-Code Comparisons



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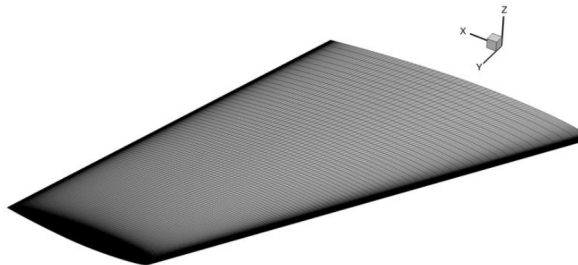


IPW3 – Onera M6 Participants

ID	Participant / Organization	Color
001	CIRA	
002	AeroTex	
003	NRC	
004	ONERA	
007	Polytechnique Montréal	
010	NASA	
015	NIAR	
019	Synopsys	



Geometry and Aerodynamic Conditions



ONERA M6 L1 Surface Mesh

Geometry

Parameter	Value
Wing	ONERA M6
Root chord [m]	1.0
Semi-span [m]	1.4760
Mean aerodynamic chord [m]	0.801673
Reference area [m ²]	1.1532
Angle of attack, α [°]	2.0

Aerodynamic Conditions

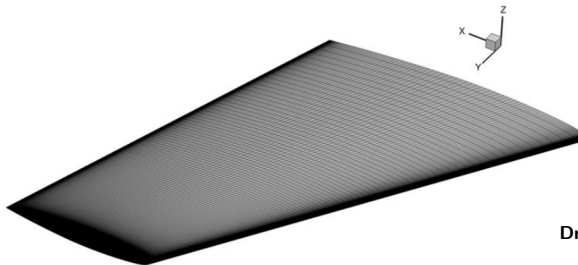
Parameter	Value
M_∞ [-]	0.40
U_∞ [m/s]	131.064
T_∞ [K]	267.15
ρ_∞ [kg/m ³]	0.68301
Re_c [-]	4.2558×10^6

Quantities of Interest

C_L , C_D , C_M , C_p



Icing Conditions



ONERA M6 L1 Surface Mesh

Grid	Nodes
L1	60.78 M
L2	7.63 M
L3	0.960 M
L4	0.122 M

Icing Conditions

Parameter	Value
T_{∞} [K]	267.15
T_0 [K]	275.7
M_{∞}	0.40
U_{∞} [m/s]	131.064
LWC [g/m ³]	0.51
MVD [μ m]	20
Exposure time [s]	1350

Droplet Distribution : 1, 3, 7 & 15 bin distributions are considered

Common Modeling Parameters

A dimensional roughness of $k_s = 1.0$ mm is prescribed as the baseline, and a common ice density of $\rho_{\text{ice}} = 680$ kg/m³ is used for the comparison.

Quantities of Interest

β , HTC , T_s , FF , m_{water} , m_{ice} , m_{evap}



ONERA M6 – Modeling Choices Across Submissions

Model / Method	001	002	004	007	019
Turbulence	$k-\omega$ TNT	$k-\omega$ SST	$k-\omega$ SST	SA	$k-\omega$ SST
Droplet formulation	Eul.	Lag.	Eul.	Eul.	Eul.
HTC method	–	BL Int.	T_{rec}	T_{rec}	–
Thermodynamics	Mess.	Mess.	Mess.	Mess.	SWIM+
Geometry evolution	Lag./IB	Interp.	–	Lag.	Lag./remesh
No. of shots	1 / Multi	1	1	1	Multi

Eul. = Eulerian; *Lag.* = Lagrangian; *SA* = Spalart–Allmaras; *BL Int.* = boundary-layer integral; T_{rec} = recovery temperature; *Mess.* = Messinger; *SWIM* = shallow-water icing model; *IB* = immersed boundary; *ALE* = arbitrary Lagrangian–Eulerian; *Interp.* = interpolation; *disp.* = displacement; *Multi* = multilayer/multishot; – = not reported.

Overall Trends

- $k-\omega$ SST and Eulerian droplet formulations are the predominant choices among the disclosed submissions.
- Thermodynamic modeling is predominantly Messinger-based, while geometry-evolution methods are more diverse.
- Both single-shot and multishot strategies are represented.



ONERA M6 – Summary of Data Received

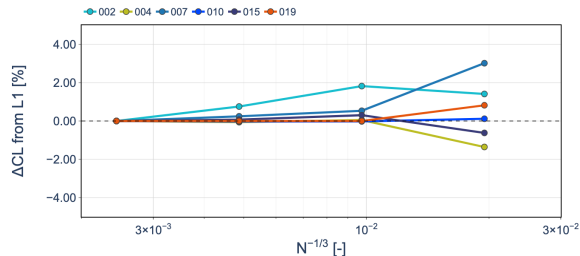
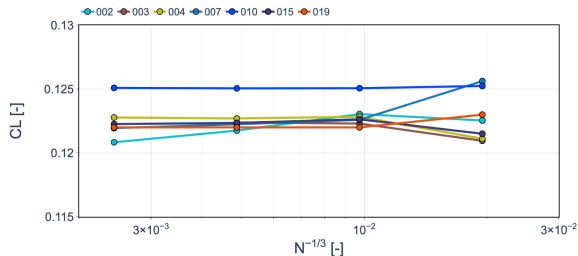
ID	L1				L2				L3				L4			
	1	3	7	15	1	3	7	15	1	3	7	15	1	3	7	15
001	✓				✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
002	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
003							*				*				*	
004	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
007	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
010	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
015	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
019	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Total	7	6	6	6	7	7	7	7	7	7	7	7	7	7	7	7

✓: populated CutData received for the corresponding droplet-bin distribution.

*: Excel results only, no CutData slices.



ONERA M6 – Lift Coefficient Grid Convergence



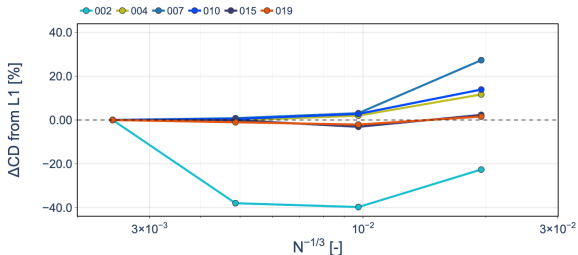
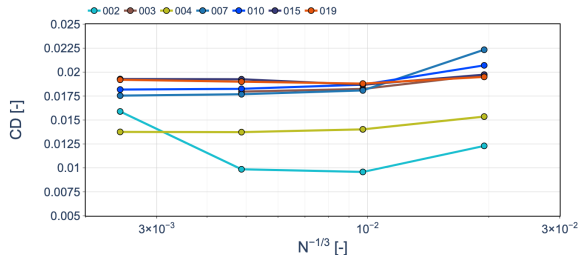
Observations

- At L1, the median lift coefficient is $C_L = 0.12213$, with an IQR of 6.89×10^{-4} (0.4956%).
- Coarse-grid deviations generally remain within a few percent of the corresponding L1 predictions.
- The code-to-code spread in C_L remains small across the grid levels.

IQR = interquartile range (75th percentile – 25th percentile).



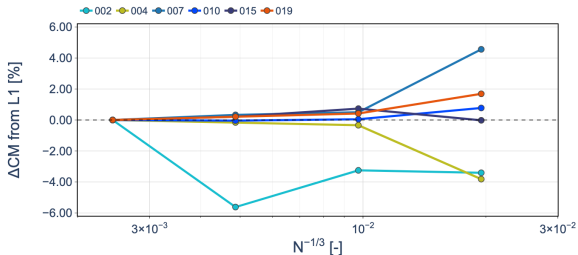
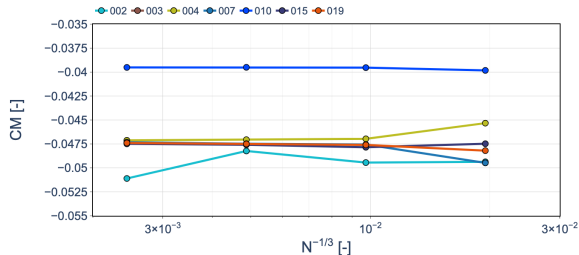
ONERA M6 – Drag Coefficient Grid Convergence



Observations

- At L1, the median drag coefficient is $C_D = 0.01786$ (178.6 drag counts), with an IQR of 2.64×10^{-3} (26.4 drag counts, 14.8%).
- C_D exhibits considerably larger code-to-code variability than C_L .
- Several submissions show non-monotonic sensitivity to spatial grid refinement.
- Relative deviations from the corresponding L1 predictions reach approximately 40% for selected submissions.
- Flow-convergence difficulties were reported for some submissions and may contribute to the observed grid sensitivity and variability in C_D .

ONERA M6 – Pitching Moment Grid Convergence

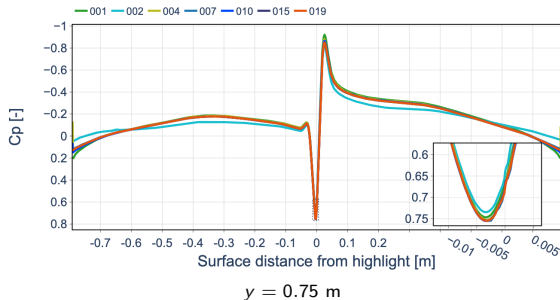


Observations

- At L1, the median pitching-moment coefficient is $C_M = -0.04736$, with an IQR of 2.93×10^{-4} (0.62%).
- Most submissions exhibit limited sensitivity to spatial grid refinement.



ONERA M6 – Surface Pressure Distribution (L1)

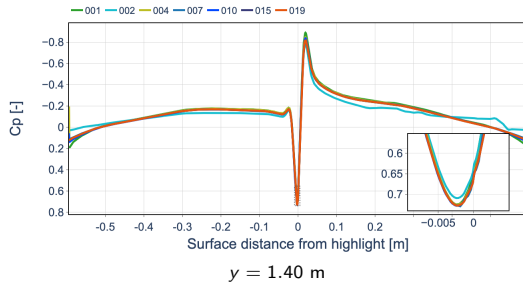
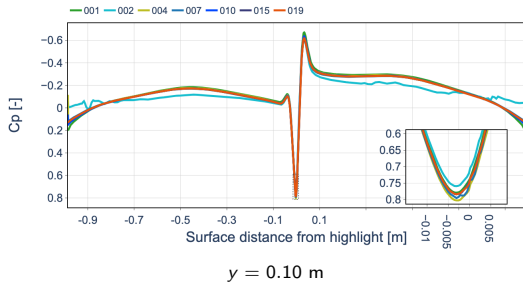


Observations

- The submitted C_p distributions are generally closely clustered across all three spanwise locations.
- The characteristic pressure variations associated with the leading-edge and suction-side peak are consistently captured by the participants.
- Overall agreement remains similar from the inboard to the outboard spanwise stations.



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ONERA M6 – Aerodynamic Assessment Summary

Grid Convergence

- C_L : very weak grid sensitivity, with closely clustered predictions across the grid hierarchy.
- C_D : substantially greater grid sensitivity and code-to-code variability, including non-monotonic behavior for selected submissions.
- C_M : limited within-code grid sensitivity for most submissions.

Code-to-Code Variability at L1

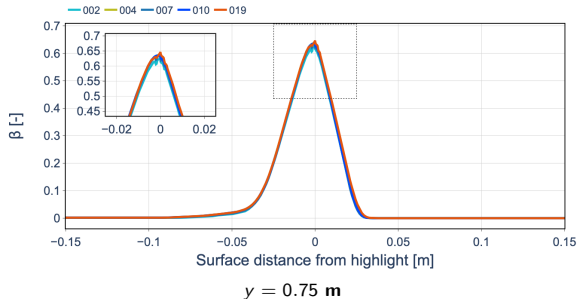
	Median	IQR	Relative IQR
C_L	0.12213	6.89×10^{-4}	0.4956%
C_D	0.01786	2.64×10^{-3}	14.8%
C_M	-0.04736	2.93×10^{-4}	0.62%

Aerodynamic Takeaway

The lift and pitching-moment predictions exhibit weak grid sensitivity and relatively tight code-to-code agreement for most submissions. In contrast, drag remains considerably more sensitive to both numerical implementation and spatial resolution. The C_p distributions are nevertheless consistently reproduced across the three spanwise stations.



ONERA M6 – Collection Efficiency (15 Bins, L1)

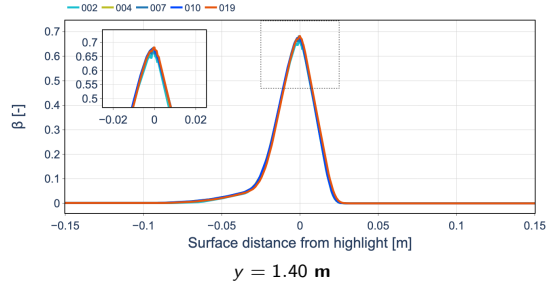
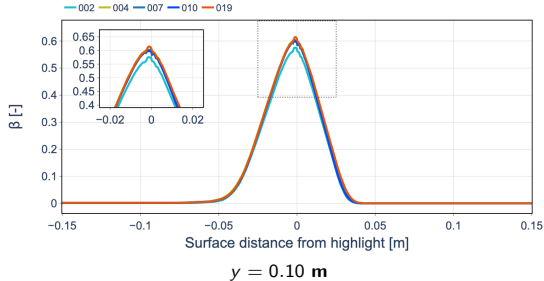


Observations

- Good code-to-code agreement is observed in the collection-efficiency distribution at the middle spanwise station ($y = 0.75 \text{ m}$).
- The peak magnitude and location are closely clustered across submissions.
- Similar agreement is observed at the inboard and outboard stations.
- Small oscillations near the peak are likely associated with the surface discretization in the highly curved leading-edge region.



ONERA M6 – Collection Efficiency (15 Bins, L1)

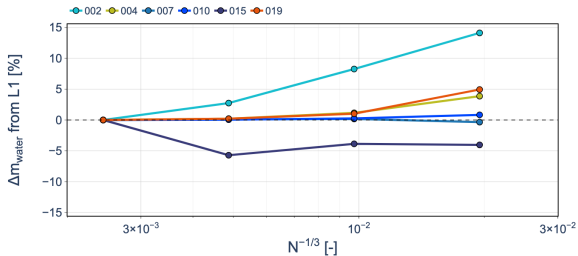
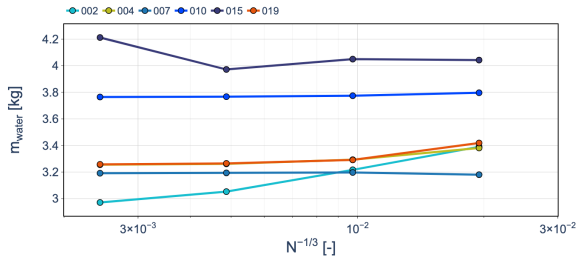


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ONERA M6 – Water Mass Grid Convergence (15 Bins)

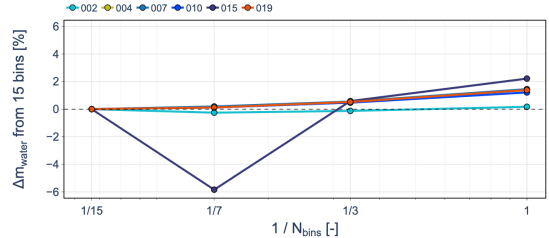
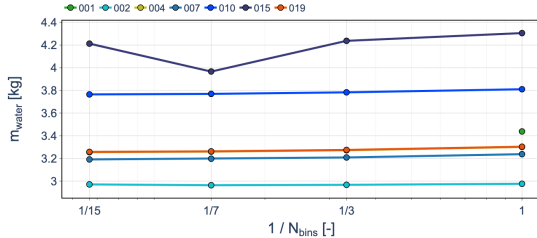


Observations

- For the 15-bin distribution, the L1 median water mass is 3.257 kg, with an IQR of 0.431 kg (13.2%).
- Most submissions exhibit limited sensitivity to spatial grid refinement, with changes generally remaining within approximately 5% of their corresponding L1 predictions.



ONERA M6 – Droplet Distribution Sensitivity (L1)

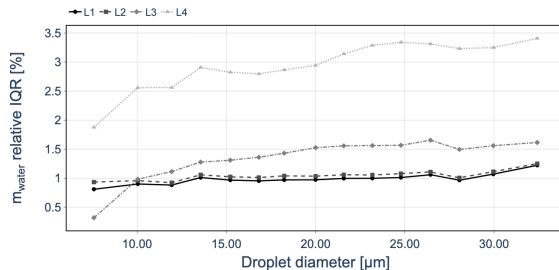
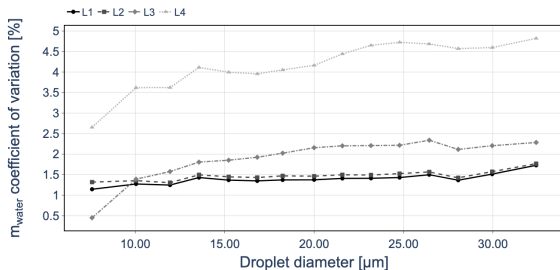


Observations

- All the distributions produce very similar impinged water masses for most submissions.
- Relative differences remain within approximately 2% for most submissions and distributions, with one isolated 7-bin deviation of approximately 6%.
- The narrow droplet-size range (6.16–33.84 μm around an MVD of 20 μm) contributes to the weak distribution sensitivity.



ONERA M6 – Water-Mass Variability by Droplet Diameter

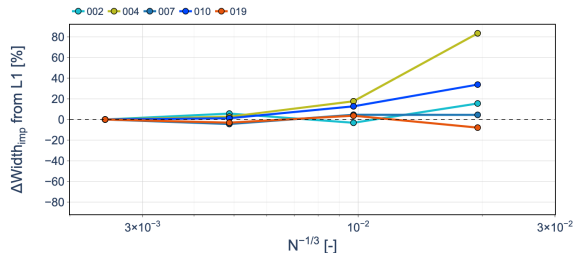
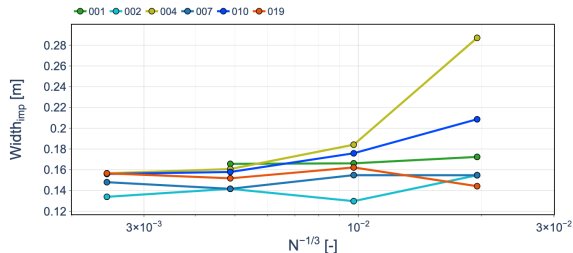


Observations

- Despite the limited dataset, variability remains low on L1/L2 ($\sim 1\%$ IQR) and increases on the coarser L3/L4 grids.
- No clear dependence on droplet diameter is observed over the relatively narrow 6.16–33.84 μm range.
- Grid resolution appears to have a stronger influence on variability than droplet size.



ONERA M6 – Impingement Width Grid Convergence (15 Bins)

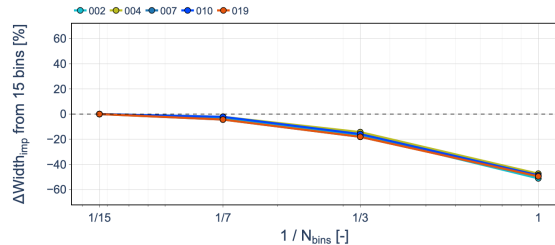
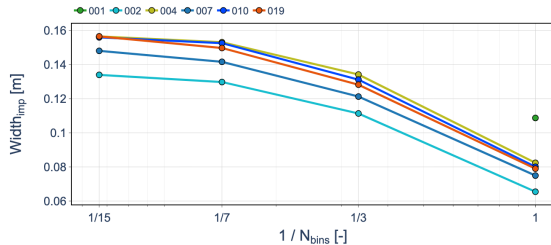


$y = 0.75 \text{ m}$

Observations

- Due to near-peak β oscillations, impingement width is retained as the primary profile-based metric, defined using the $\beta = 0.001$ threshold.
- Most submissions show limited sensitivity between L1 and L2.
- Larger, non-monotonic deviations appear on coarser grids, indicating increased sensitivity of the impingement limits to grid resolution.

ONERA M6 – Impingement Width Distribution Sensitivity (L1)



$y = 0.75 \text{ m}$

Observations

- Impingement width systematically increases from 1 to 15 bins, with noticeable differences remaining between 7 and 15 bins.
- In contrast, integrated water mass varies by only approximately 2% for most submissions.
- Droplet-distribution resolution therefore affects impingement extent more strongly than total collected water mass.



ONERA M6 – Droplet Impingement Summary

Spatial Grid Convergence

- Collection-efficiency distributions show good code-to-code agreement across the three spanwise stations.
- Integrated water mass exhibits limited grid sensitivity for most submissions, while code-to-code differences remain larger.
- Impingement limits are more sensitive to spatial resolution.

Droplet-Size Distribution

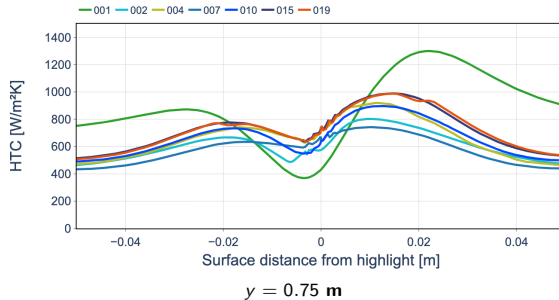
- Integrated water mass is weakly sensitive to droplet discretization; even the single-MVD approximation remains within approximately 2% of the 15-bin prediction for most submissions.
- In contrast, impingement width systematically increases from 1 to 15 bins, with convergence not yet clearly demonstrated at 15 bins.
- The relatively narrow 6.16–33.84 μm range around the 20 μm MVD contributes to the weak sensitivity of the integrated water mass.

Impingement Takeaway

Integrated water mass can appear converged while the spatial impingement extent remains sensitive to both spatial and droplet-distribution resolution.



ONERA M6 – Convective Heat-Transfer Distribution (L1)

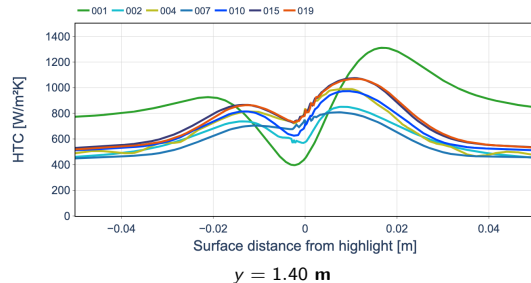
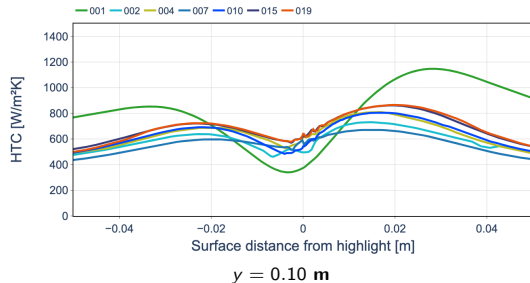


Observations

- At the prescribed 1 mm roughness, considerable code-to-code differences are observed in HTC , particularly near the leading edge.
- Similar variability is observed at the other spanwise stations, indicating that roughness alone does not explain the code-to-code scatter.
- Increasing roughness generally increases the predicted HTC , although the magnitude of the roughness effect varies between codes.



ONERA M6 – Convective Heat-Transfer Distribution (L1)

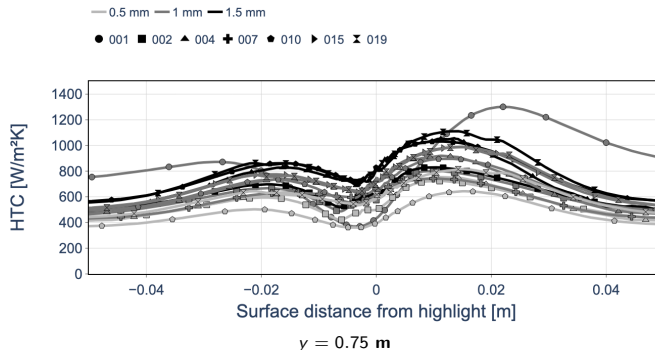


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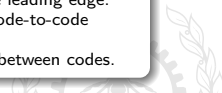


ONERA M6 – Convective Heat-Transfer Distribution (L1)

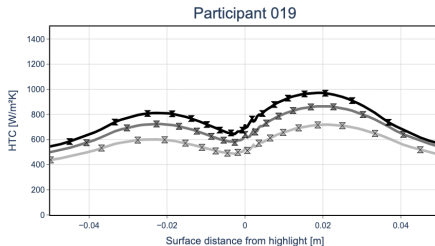
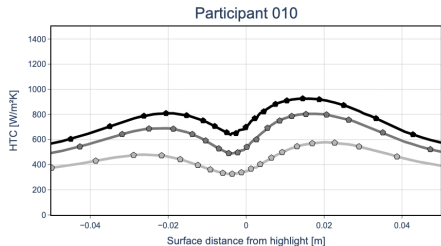
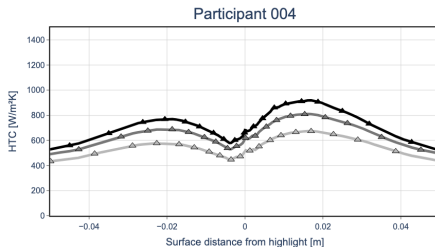
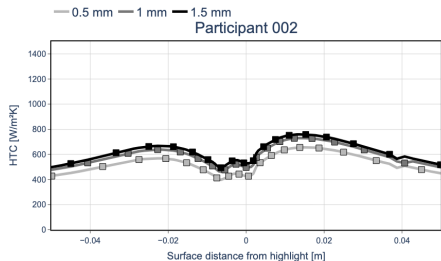


Observations

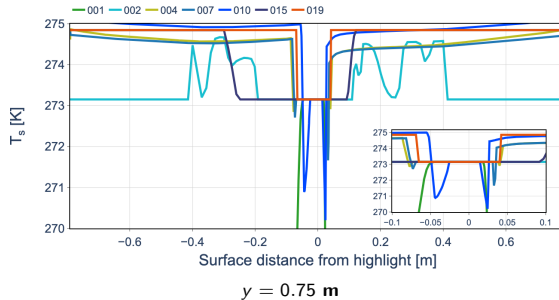
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ONERA M6 – Participant-Level HTC Roughness Sensitivity (L1)



ONERA M6 – Surface Temperature Distribution (L1)

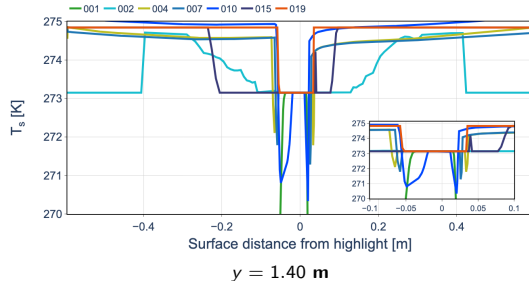
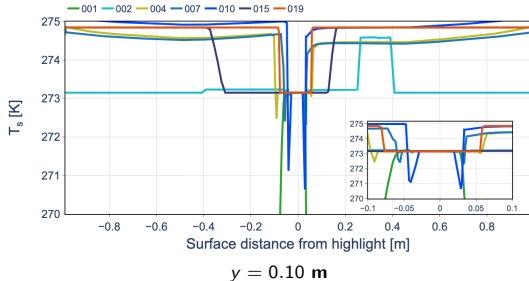


Observations

- Most submissions predict surface temperatures close to the freezing temperature over a large portion of the surface.
- Local code-to-code differences are concentrated near the leading edge.
- Similar differences are observed at the inboard and outboard spanwise stations.



ONERA M6 – Surface Temperature Distribution (L1)

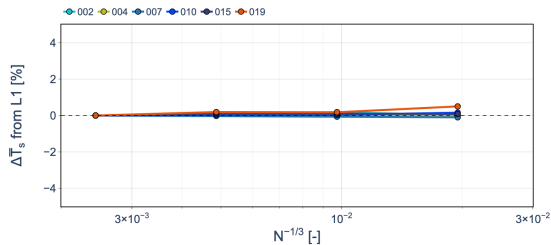
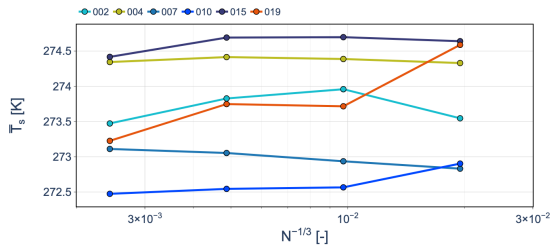


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ONERA M6 – Mean Surface Temperature Grid Convergence



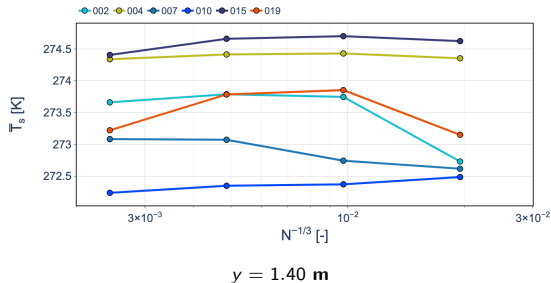
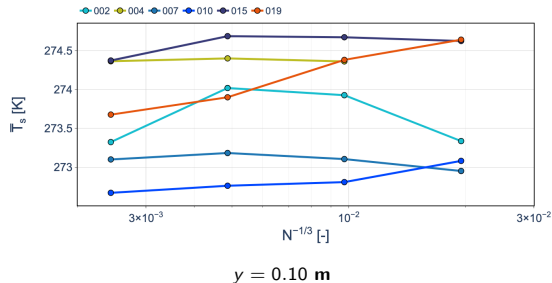
$y = 0.75 \text{ m}$

Observations

- Mean T_s is area-weighted (ds) over the iced surface ($h_{ice} > 0$).
- Grid sensitivity is limited ($< 1\%$ for most submissions), with closely clustered predictions.
- Similar behavior is observed across the spanwise stations.
- Roughness changes mean T_s by approximately $\pm 1 \text{ K}$, with a code-dependent response.



ONERA M6 – Mean Surface Temperature Grid Convergence

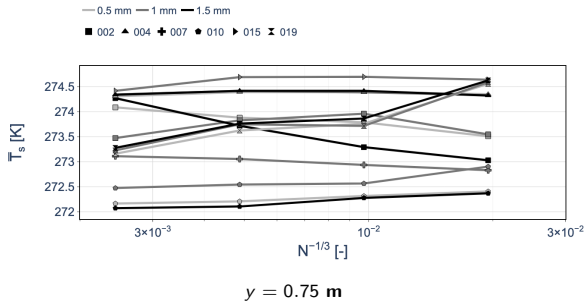


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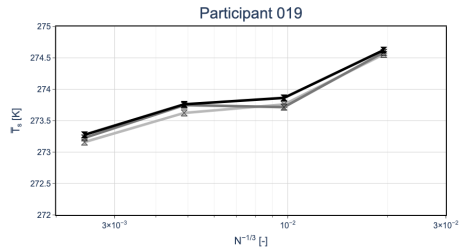
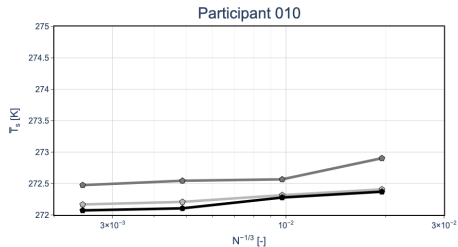
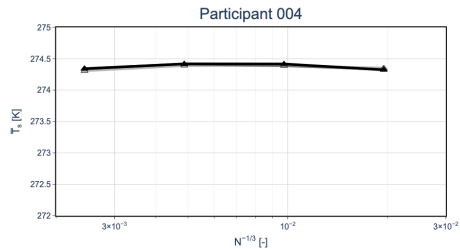
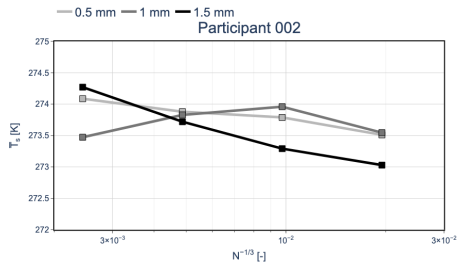


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ONERA M6 – Participant-Level Mean Surface Temperature Roughness Sensitivity (L1)



ONERA M6 – Integrated Convective Heat Transfer

Grid-Convergence Metric

To assess whether the differences in the local HTC distributions translate into a clear grid-convergence behavior, an integrated convective heat-transfer quantity is evaluated along the surface slice:

$$Q'_c = \int HTC(s) [T_s(s) - T_{rec}(s)] ds \quad [Q'_c] = W/m.$$

The submitted HTC and surface temperature T_s are used directly. The recovery temperature T_{rec} , however, was not part of the submitted dataset and must therefore be reconstructed.

Recovery Temperature Reconstruction

The submitted pressure coefficient is used to estimate the local external-flow state and subsequently the recovery temperature:

$$C_p(s) \longrightarrow p_e(s) \longrightarrow M_e(s) \longrightarrow T_e(s) \longrightarrow T_{rec}(s)$$

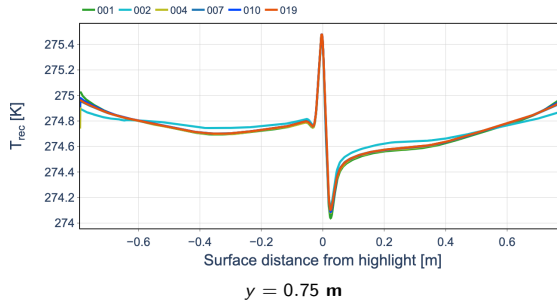
Assuming $p_s \simeq p_e$ and an isentropic external flow,

$$\frac{p_e}{p_\infty} = 1 + \frac{\gamma}{2} M_\infty^2 C_{p,e}, \quad M_e^2 = \frac{2}{\gamma - 1} \left[\left(\frac{p_0}{p_e} \right)^{(\gamma-1)/\gamma} - 1 \right],$$

$$T_e = \frac{T_0}{1 + \frac{\gamma-1}{2} M_e^2}, \quad T_{rec} = T_e \left[1 + r \frac{\gamma-1}{2} M_e^2 \right], \quad r = Pr^{1/3} \approx 0.9.$$



ONERA M6 – Reconstructed Recovery Temperature Distribution (L1)

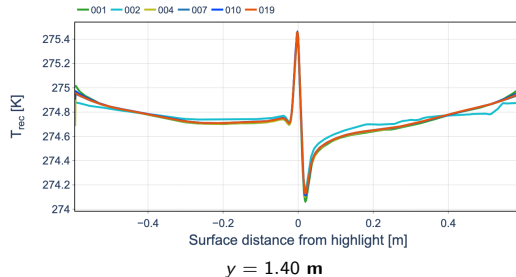
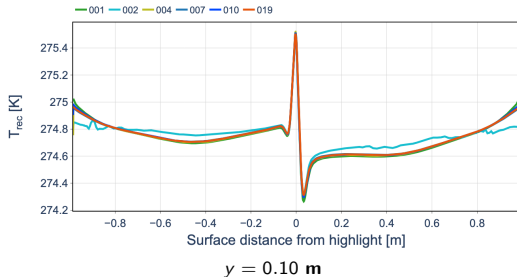


Observations

- The reconstructed recovery-temperature distributions are closely aligned across submissions.
- This agreement is consistent with the good code-to-code agreement previously observed for C_p , from which T_{rec} is reconstructed.
- Similar agreement is observed at the inboard and outboard spanwise stations.



ONERA M6 – Reconstructed Recovery Temperature Distribution (L1)

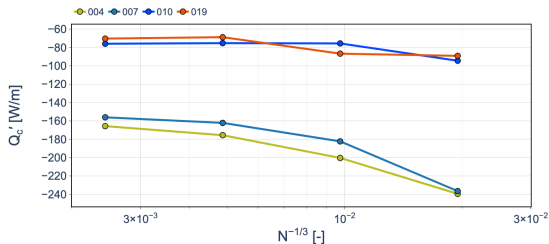
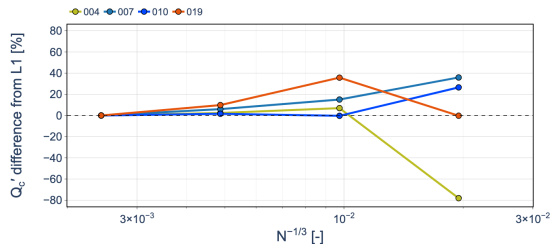


Observations

- The reconstructed recovery-temperature distributions are closely aligned across submissions.
- This agreement is consistent with the good code-to-code agreement previously observed for C_p , from which T_{rec} is reconstructed.
- Similar agreement is observed at the inboard and outboard spanwise stations.



ONERA M6 – Convective Heat Transfer Grid Convergence

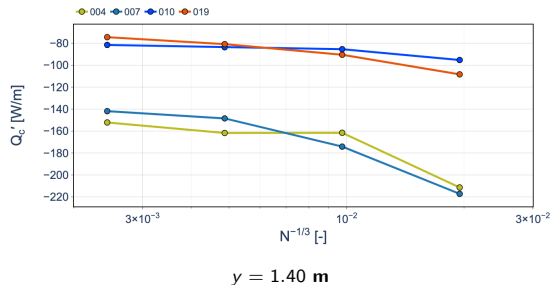
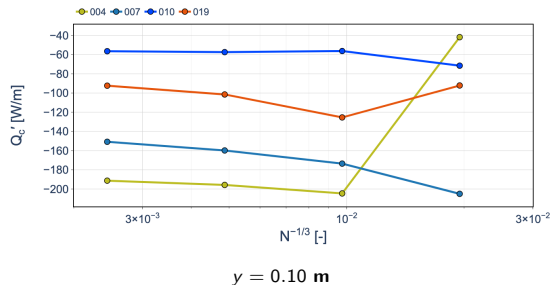
 $y = 0.75 \text{ m}$ 

Observations

- Considerable code-to-code variability is observed in Q'_c , exceeding the L1–L2 grid effect for most submissions.
- Large variability persists across all spanwise stations.
- Increasing surface roughness systematically increases $|Q'_c|$.



ONERA M6 – Convective Heat Transfer Grid Convergence

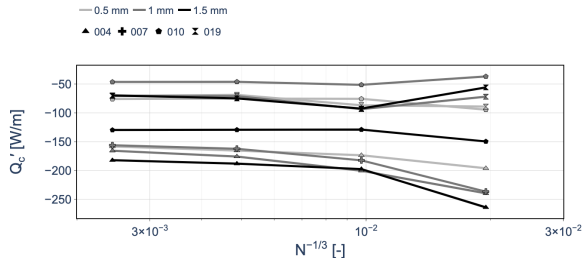


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- Increasing surface roughness systematically increases $|Q'_c|$.



ONERA M6 – Convective Heat Transfer Grid Convergence



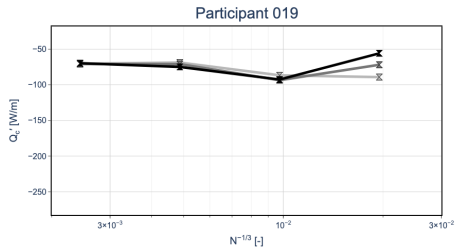
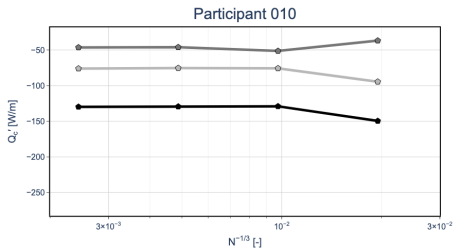
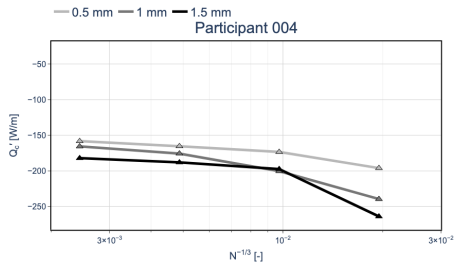
$y = 0.75$ m- Roughness Sensitivity

Observations

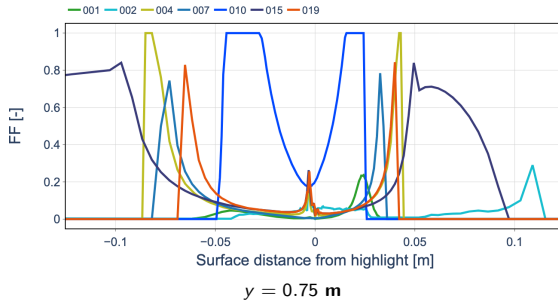
- Considerable code-to-code variability is observed in Q'_c , exceeding the L1–L2 grid effect for most submissions.
- Large variability persists across all spanwise stations.
- Increasing surface roughness systematically increases $|Q'_c|$.



ONERA M6 – Participant-Level HTC Roughness Sensitivity (L1)



ONERA M6 – Freezing-Fraction Distribution (L1)

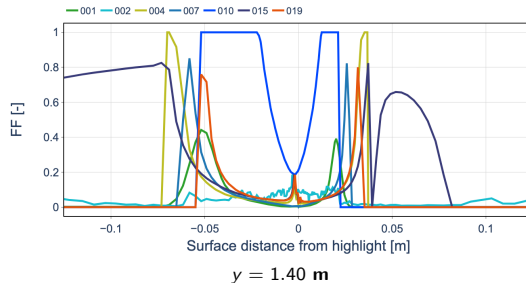
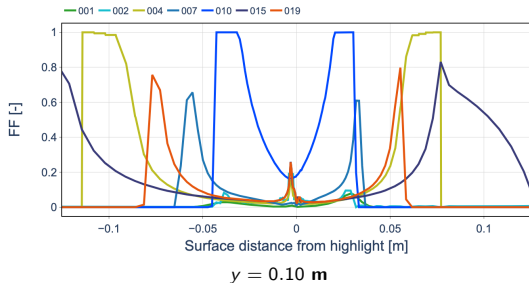


Observations

- Large code-to-code differences are observed in both the magnitude and spatial extent of the predicted freezing fraction.
- Substantial differences in the predicted freezing regime persist across the inboard and outboard spanwise stations.
- Roughness affects the extent of the ice-covered region, with both the magnitude and direction of the response varying between codes.



ONERA M6 – Freezing-Fraction Distribution (L1)

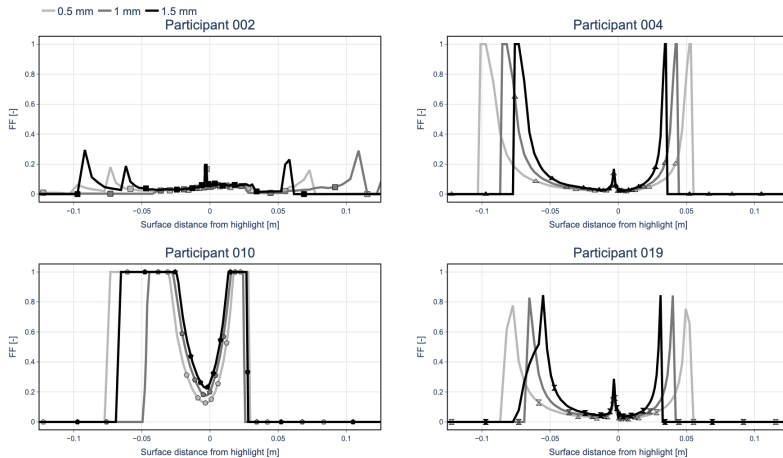


Observations

- Large code-to-code differences are observed in both the magnitude and spatial extent of the predicted freezing fraction.
- Substantial differences in the predicted freezing regime persist across the inboard and outboard spanwise stations.
- Roughness affects the extent of the ice-covered region, with both the magnitude and direction of the response varying between codes.



ONERA M6 – Participant-Level Freezing Fraction Roughness Sensitivity (L1)



$y = 0.75 \text{ m}$ – Individual Participants



ONERA M6 – Convective Heat Transfer & Thermodynamics Summary

Convective Heat Transfer

- Considerable code-to-code variability is observed in HTC , while reconstructed T_{rec} distributions remain closely aligned.
- Consequently, Q'_c exhibits substantial code-to-code variability, with an L1 IQR of 83.9 W/m at $y = 0.75$ m.

Surface Thermodynamics

- Mean surface temperature shows limited grid sensitivity and comparatively close agreement for most submissions.
- In contrast, freezing fraction exhibits larger code-to-code variability in both magnitude and spatial extent.

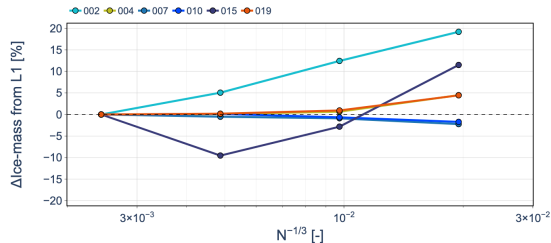
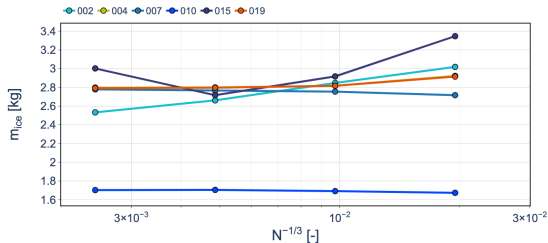
Roughness Sensitivity

- Increasing roughness consistently increases HTC and $|Q'_c|$.
- Mean surface temperature and freezing fraction exhibit code-dependent responses to roughness.

Thermodynamics Takeaway

Despite consistent aerodynamic loading and T_{rec} , substantial variability develops through the convective and freezing models, with the largest differences appearing in the predicted freezing regime.

ONERA M6 – Ice Mass Grid Convergence (15 Bins)

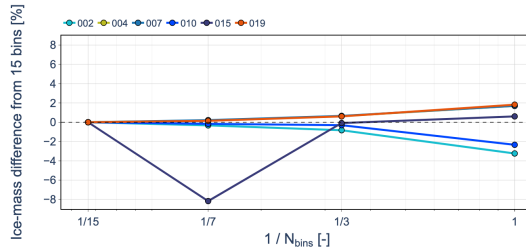
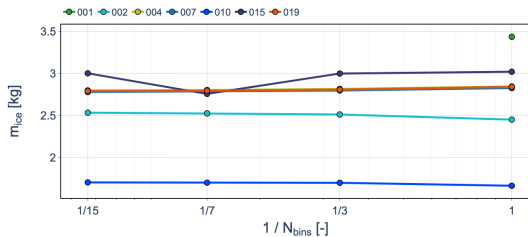


Observations

- At L1, median ice mass is 2.79 kg with a 7.2% IQR.
- Grid sensitivity is submission-dependent, with L4 deviations ranging from below 5% to approximately 20%.



ONERA M6 – Ice Mass Sensitivity to Droplet Distribution

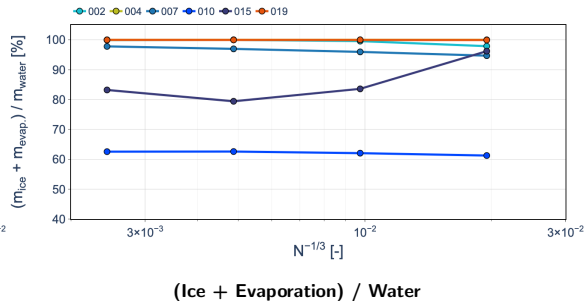
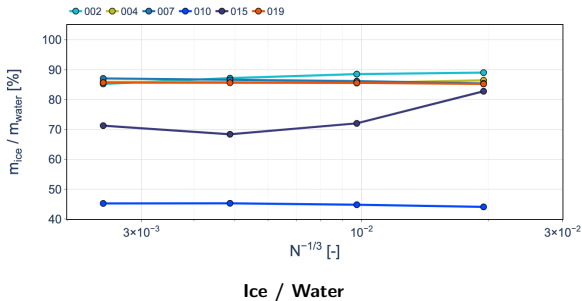


Observations

- Ice mass shows limited sensitivity to droplet-distribution resolution, generally within $\sim 3\%$ of the 15-bin prediction.
- This is consistent with the weak distribution sensitivity observed for impinged water mass.



ONERA M6 – Ice and Evaporation Mass Ratios

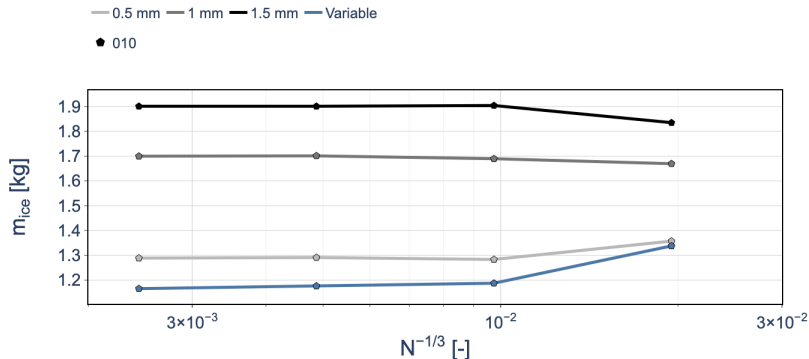


Observations

- Most submissions convert approximately 85–89% of the impinging water mass into ice.
- Accounting for evaporation brings the mass balance close to 100% for some submissions.
- The remaining mass deficit for some submissions may be associated with water shedding from the surface or other mechanisms.



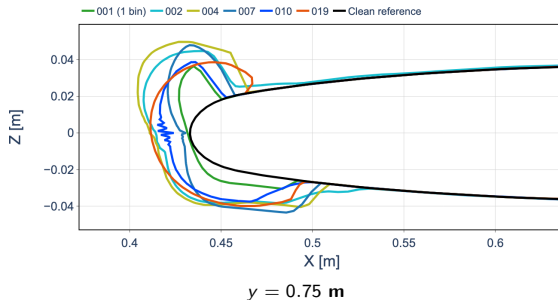
ONERA M6 – Ice Mass Roughness Sensitivity



Observations

- One submission provided additional results for multiple prescribed roughness heights.
- Ice mass increases substantially with roughness, from approximately 1.3 kg at $k_s = 0.5$ mm to 1.9 kg at $k_s = 1.5$ mm on L1.

ONERA M6 – Ice Shapes (L1, 15 Bins)

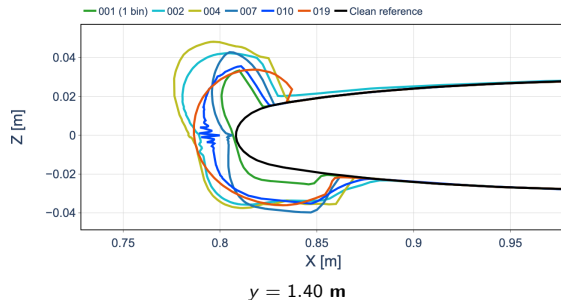
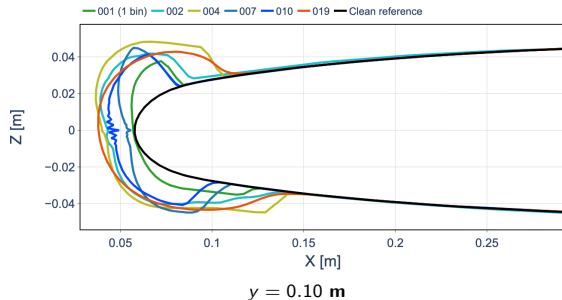


Observations

- Considerable code-to-code variability is observed in ice shape, particularly in horn geometry and local thickness.
- Similar variability persists at the inboard and outboard spanwise locations.
- Shape variability is consistent with the thermodynamic spread, despite comparatively good impingement agreement.



ONERA M6 – Ice Shapes (L1, 15 Bins)

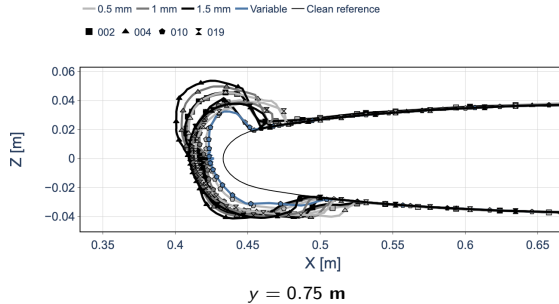


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- Similar variability persists at the inboard and outboard spanwise locations.
- Shape variability is consistent with the thermodynamic spread, despite comparatively good impingement agreement.



ONERA M6 – Ice-Shape Sensitivity to Surface Roughness

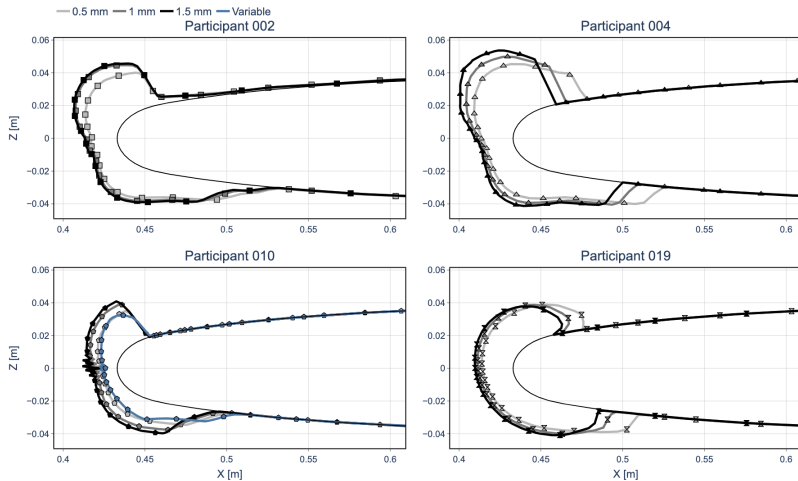


Observations

- Surface roughness affects ice geometry, horn height, extent, and orientation.
- The magnitude of the roughness effect is participant-dependent.
- Roughness alone does not explain the overall code-to-code variability.



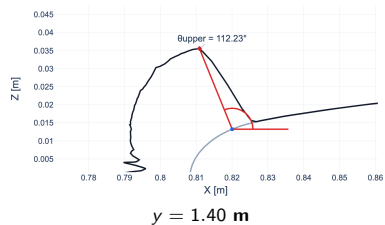
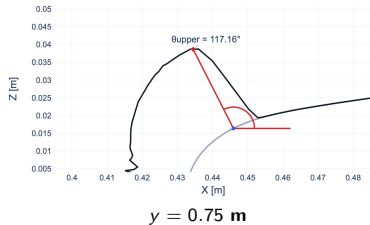
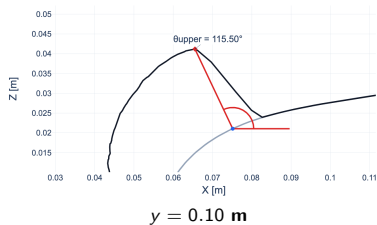
ONERA M6 – Ice-Shape Sensitivity to Surface Roughness



$y = 0.75 \text{ m}$ – Individual Participants



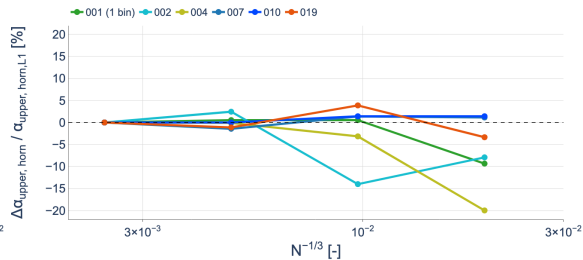
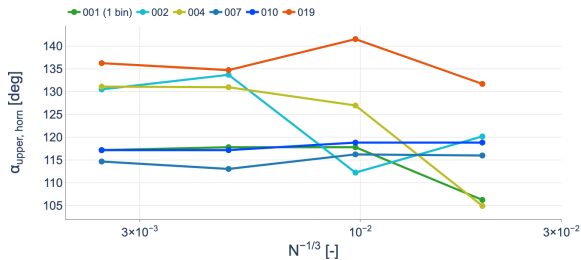
ONERA M6 – Upper-Horn Angle Extraction



Definition

- The upper-horn angle is evaluated independently at each of the three spanwise slices.



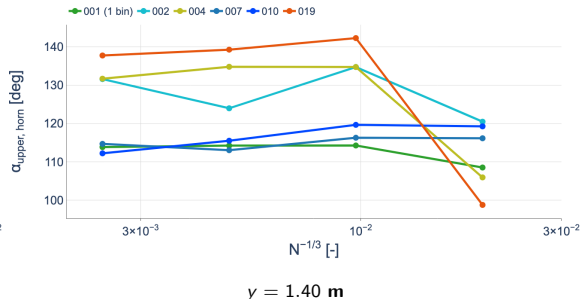
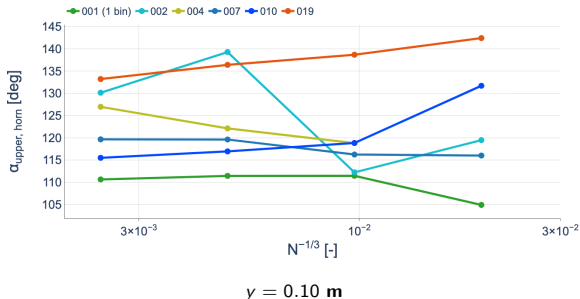
ONERA M6 – Upper-Horn Angle Variation ($y = 0.75$ m)

Observations

- Substantial code-to-code variability is observed in the upper-horn angle at L1.
- Refinement does not lead to convergence toward a common horn-angle prediction.



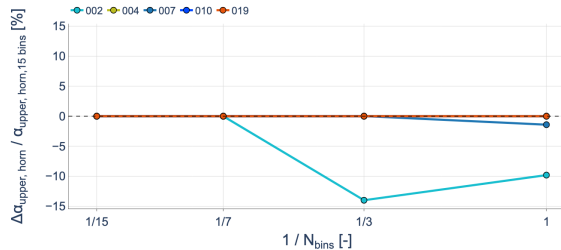
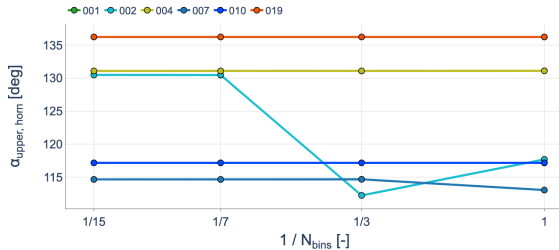
ONERA M6 – Upper-Horn Angle Variation at Other Slices



Observations

- Similar horn-angle variability is observed at the inboard and outboard slices.
- The spread remains larger than the grid-refinement effect for several submissions.



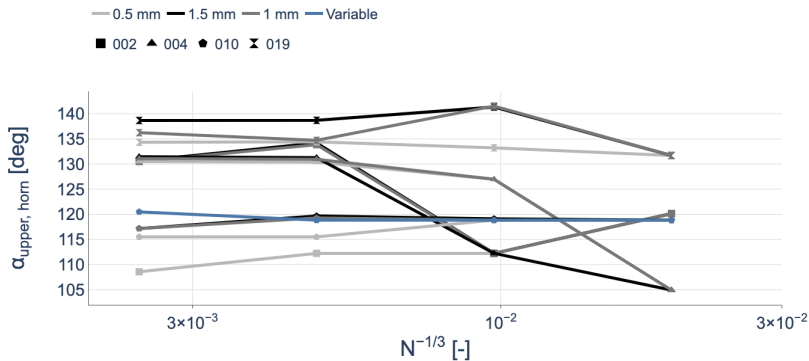
ONERA M6 – Upper-Horn Angle Distribution Sensitivity ($y = 0.75$ m)

Observations

- The upper-horn angle is insensitive to droplet-distribution resolution for most submissions.
- One submission exhibits a stronger sensitivity when reducing the number of bins, reaching approximately 14% relative to the 15-bin prediction.



ONERA M6 – Upper-Horn Angle Sensitivity to Roughness

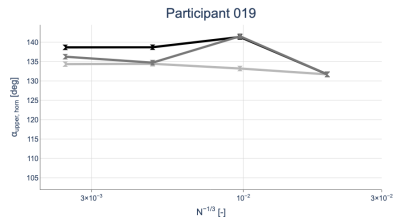
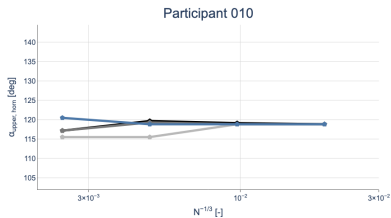
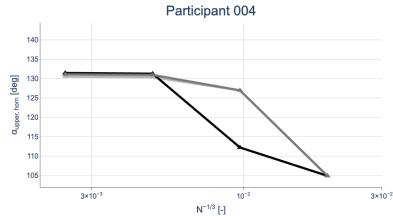
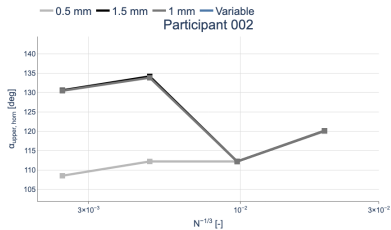


Observations

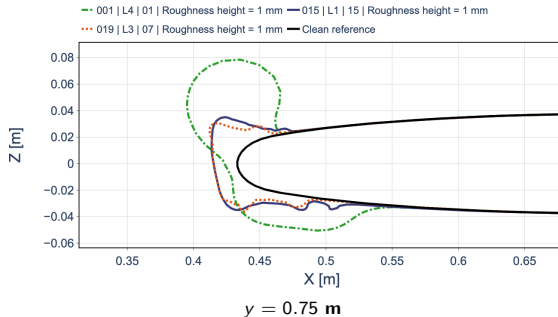
- Surface roughness affects the upper-horn angle, with a participant-dependent response.



ONERA M6 – Upper-Horn Angle Sensitivity to Roughness



ONERA M6 – Multilayer Ice-Shape Comparison

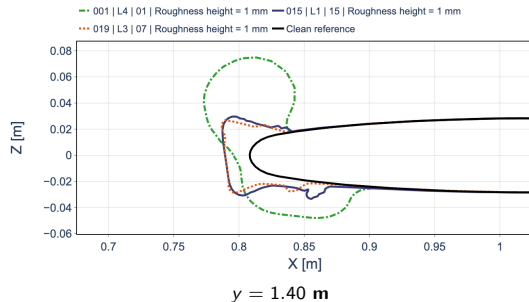
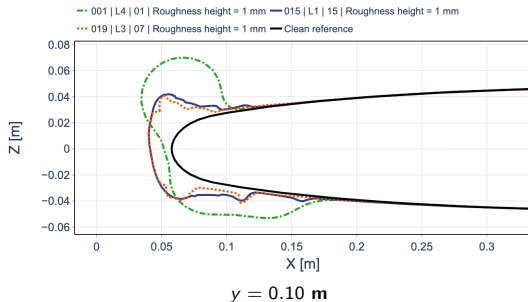


Observations

- Considerable code-to-code variability is observed in horn geometry and local ice thickness.
- Using multiple accretion layers does not lead to convergence toward a common ice shape.
- Similar variability persists at the inboard and outboard spanwise stations.



ONERA M6 – Multilayer Ice-Shape Comparison



Observations

- Considerable code-to-code variability is observed in horn geometry and local ice thickness.
- Using multiple accretion layers does not lead to convergence toward a common ice shape.
- Similar variability persists at the inboard and outboard spanwise stations.



ONERA M6 – Ice Accretion Summary

Ice Mass

- Grid sensitivity is submission-dependent, while sensitivity to droplet discretization remains limited.
- Code-to-code differences remain substantial and generally exceed the grid and droplet-discretization effects.

Mass Partition

- Most submissions convert approximately 85–89% of the impinged water mass into ice.
- Evaporation closes much of the remaining mass balance; lower ratios for some submissions are consistent with reported water shedding.

Ice Geometry

- Considerable code-to-code variability is observed in horn height, orientation, and local ice thickness.
- Similar geometric variability persists across the three spanwise sections.

Ice Accretion Takeaway

Good agreement in water impingement does not translate into comparable ice accretion; substantial differences emerge in ice mass and final ice geometry, highlighting the influence of the thermodynamic treatment.

ONERA M6 – Overall Assessment

Overall Assessment

- **Aerodynamics:** Limited grid and code-to-code variability overall, except for C_D .
- **Droplet Impingement:** Good overall agreement; impingement extent is more sensitive to spatial and droplet-distribution resolution than integrated water mass.
- **HTC & Thermodynamics:** Substantial code-to-code variability, particularly in HTC , convective heat transfer, and freezing fraction.
- **Ice Accretion:** Comparable water impingement does not translate into comparable ice mass or final ice geometry.

Overall Takeaway

Code-to-code variability increases through the icing simulation chain, with large differences emerging in thermodynamics and final ice accretion. HTC and thermodynamic modeling deserve closer investigation for these glaze conditions, while geometry-evolution methods also clearly influence the final ice shape, even when thermodynamic predictions are comparable.

